Tight instances of the Lonely Runner Conjecture: one-entry modifications and the growth of speeds

Yuhan Zhang

Yangzhou University, Yangzhou, China
yuhanzh6@gmail.com

Abstract

For a set V of $n-1$ distinct positive integers let $\mathrm{LR}(V) = \max_t \min_{v\in V} \|vt\|$, where $\|x\|$ is the distance from x to the nearest integer. The Lonely Runner Conjecture asserts $\mathrm{LR}(V) \geq 1/n$, and V is called *tight* if equality holds. Goddyn and Wong determined which of the sets obtained from $[n-1] = \{1, \ldots, n-1\}$ by multiplying a single entry by an integer are tight. We complete the classification of arbitrary one-entry modifications $([n-1]\setminus\{r\})\cup\{w\}$, $w \geq n$: for $n \geq 4$ such a set is tight if and only if it is of Goddyn–Wong type or is one of the two sporadic sets $\{1,3,4,7\}$ and $\{1,3,4,5,9\}$ found by Wills. The proof is elementary and rests on an exact description of the region left uncovered by the retained speeds. We then show that the largest speed $M(n)$ occurring in a tight one-entry modification satisfies $M(n) < 0.60\, n\log n + 52\, n$ and $\limsup_{n\to\infty} M(n)/(n\log n) = \frac{1}{2}$; in particular, tight sets are not confined to speeds $O(n)$. Along the way we give a short proof of Liu's closed formula for $\mathrm{LR}([n-1]\setminus\{2\})$ and show that $\mathrm{LR}([n-1]\setminus\{r\}) > 1/(n-1)$ for every $r \leq n-2-2\sqrt{n}$, extending the range $r \leq (n-1)/2$ covered by Liu's theorem. For two-entry modifications we show that no tight instance removes a speed q with $2 \leq q \leq n/10$ once $n \geq \max(40, 10q)$, whatever the second removed speed and the inserted speeds are; an exact-arithmetic census for $8 \leq n \leq 140$ with inserted speeds at most $8n$ finds only two tight two-entry modifications, both already listed by Goddyn and Wong.

1 Introduction

1.1 Background

Let $\|x\|$ denote the distance from $x \in \mathbb{R}$ to the nearest integer. For a finite set $V \subset \mathbb{N}$ of *speeds* put

$$\mathrm{LR}(V) \;=\; \max_{t\in[0,1)} \; \min_{v\in V} \|vt\|\,;$$

the maximum exists because $t \mapsto \min_v \|vt\|$ is continuous on the circle $\mathbb{R}/\mathbb{Z}$. The *Lonely Runner Conjecture* of Wills [33] and Cusick [13] asserts that $\mathrm{LR}(V) \geq \frac{1}{n}$ whenever $|V| = n-1$. In the usual picture, n runners with distinct constant speeds run around a circular track of length one, and V records the speeds relative to one distinguished runner; we keep this convention throughout, so that n always denotes the number of runners and V has $n-1$ elements.

The conjecture is known for $n \leq 7$ [7, 15, 8, 11, 3] and, in published form, for $n \in \{9, 10\}$ [32]; the cases $n = 8$ [27], $n \in \{11, 12, 13\}$ [30] and $n \in \{14, 15\}$ [2] have been announced in computer-assisted preprints (see also [28] for an independent treatment of $n = 9$). We refer to the survey of Perarnau and Serra [24] for the history of the problem. For general n, Bedert

[6] recently showed that k speeds always admit loneliness at least $\frac{1}{2k} + \frac{1}{k^{5/3+o(1)}}$, a polynomial improvement over Tao's bound $\frac{1}{2k} + \frac{(\log k)^{1-o(1)}}{k^2}$ [31].

This paper does not address the conjecture itself. It is concerned with the structure of its *equality case*.

1.2 Tight instances

By Dirichlet's approximation theorem, $\mathrm{LR}([n-1]) = \frac{1}{n}$ for every n, so the conjectured bound is best possible. A set V of $n-1$ speeds with $\mathrm{LR}(V) = \frac{1}{n}$ is called *tight*; dilations kV of tight sets are tight. For at most five speeds the tight sets are known completely: up to dilation they are the baseline $[n-1]$ and, for four and five speeds respectively, the sets

$$T_1 = \{1,3,4,7\} \quad (n=5), \qquad T_2 = \{1,3,4,5,9\} \quad (n=6),$$

found by Wills [34] after a comment of P. Flor; see [13, 14] (at most three speeds), [12] (four speeds) and [11, Thm. 3] (five speeds), and the account in [24, §4]. Wills also found the tight set $\{1,2,3,4,5,7,12\}$ with $n=8$. Goddyn and Wong [18] recognised the latter as the first member of an infinite family obtained by *accelerating* one runner of the baseline, and listed further tight sets such as $\{1,4,5,6,7,11,13\}$. In the open-problem list of their survey, Perarnau and Serra state as Problem 1 the characterisation of tight instances, noting that it "has seen no further progress since the work of Goddyn and Wong" [24, §11, Problem 1].

Related work concerns sets that are *almost* tight. Kravitz [21] conjectured that the loneliness of $n-1$ speeds is either at least $\frac{1}{n-1}$ or of the form $\frac{\sigma}{(n-1)\sigma+1}$ with $\sigma \in \mathbb{N}$; Fan and Sun [16] disproved this for four speeds and proposed an amended version. Giri and Kravitz [17] showed that the accumulation points of the m-th lonely runner spectrum form precisely the $(m-1)$-st spectrum. Beck and Everett [4] proved that every tight instance (and every counterexample) with k speeds satisfies a nontrivial integer linear relation of ℓ^1-norm at most $\min\bigl(2k+3, \frac{k+1}{k-1}\,\mathrm{flt}(k)\bigr)$, where $\mathrm{flt}(k)$ is the flatness constant. In the opposite direction, Bohman and Peng [10] and Pomerance [26] showed that a set of k speeds whose largest element lies in $(k,\, 2k - c\log^2 k)$ is never tight, while tight sets with largest speed $2k - \Theta(\log k)$ arise from the Goddyn–Wong criterion and a construction of Erdős for the Jacobsthal function [24, §4].

1.3 One-entry modifications

For $r \in [n-1]$ and an integer w write

$$[n-1]_{r\to w} \;=\; \bigl([n-1]\setminus\{r\}\bigr)\cup\{w\},$$

and $[n-1]_{r\mapsto mr}$ in the special case $w = mr$, $m \in \mathbb{N}$. Since $[n-1]_{r\to w}$ has $n-1$ elements only if $w \notin [n-1]\setminus\{r\}$, and $w = r$ returns the baseline, we always assume $w \geq n$.

Theorem 1.1 (Goddyn–Wong [18, Thm. 2.3])**.** *Let $n \geq 2$, $r \in [n-1]$ and $m \geq 1$, and put $s = n - r$. Then $\mathrm{LR}([n-1]_{r\mapsto mr}) \geq \frac{1}{n}$, with equality if and only if $n = 2$, or $(n,r,m) = (3,1,4)$, or*

$$\gcd(r,b) > 1 \qquad \textit{for every integer } b \textit{ with } s \leq b \leq ms-1. \tag{1}$$

Goddyn and Wong also proved (Theorem 2.4) that, for fixed r, only finitely many replacements $r \to w$ with $r \nmid w$ can be tight, and remarked on page 7 that this "partially explains why there are no further sporadic speed sets resembling T_1 and T_2 where the speed 2 is replaced by an odd integer" [18, p. 7]. Our first result settles the one-entry case completely.

Theorem 1.2 (Classification of one-entry modifications)**.** *Let $n \geq 4$, $r \in [n-1]$ and $w \geq n$, and put $s = n - r$. Then $[n-1]_{r\to w}$ is tight if and only if one of the following holds:*

(i) *$r \mid w$ and (1) holds with $m = w/r$;*

(ii) *$(n, r, w) \in \{(5, 2, 7), (6, 2, 9)\}$, i.e. the set is T_1 or T_2.*

Remark 1.3. (a) Condition (i) forces $2r > n - 1$. Indeed, if $2r \leq n - 1$ then $s \geq r + 1$, and since $m \geq 2$ the interval $[s, ms - 1]$ contains at least $s \geq r$ consecutive integers, hence an integer $b \equiv 1 \pmod r$, for which $\gcd(r, b) = 1$. Consequently *no tight one-entry modification removes a speed r with $3 \leq r \leq (n-1)/2$*, and the speed 2 is the only removable speed in this range, through T_1 and T_2.

(b) For $n \leq 3$ the situation is elementary. For $n = 2$ every set $\{w\}$ is tight. For $n = 3$ and $r = 1$ Theorem 1.1 applies, and the only tight example is $(n, r, w) = (3, 1, 4)$. For $n = 3$ and $r = 2$, an even w is covered by Theorem 1.1 (and gives no tight set), while for odd w all speeds of $\{1, w\}$ are odd, so $t = \frac{1}{2}$ gives $\mathrm{LR}(\{1, w\}) = \frac{1}{2} > \frac{1}{3}$.

The proof of Theorem 1.2 is based on the *uncovered region*

$$U(n, r) = \Big\{ t \in [0, 1) : \ \|it\| > \tfrac{1}{n} \text{ for all } i \in [n-1] \setminus \{r\} \Big\},$$

which the inserted speed w alone must cover if $[n-1]_{r\to w}$ is tight. We determine $U(n, r)$ exactly (Propositions 2.5 and 2.6). For $2r > n - 1$ this forces $r \mid w$, and Theorem 1.1 applies. For $3 \leq r \leq (n-1)/2$ the longest component of $U(n, r)$ limits w, and a short argument with the Jacobsthal function shows that the two components next to $\frac{1}{r}$ then impose incompatible conditions on $w \bmod r$. For $r = 2$ the two components of $U(n, 2)$ around $\frac{1}{2}$ leave exactly four candidates; two of them are T_1 and T_2, and the other two are excluded by a single explicit point.

Our second result concerns the size of the inserted speed.

Theorem 1.4 (Extremal speed)**.** *For $n \geq 3$ let $M(n)$ be the largest w such that $[n-1]_{r\to w}$ is tight for some $r \in [n-1]$ and $w \geq n$ (with $M(n) = -\infty$ if there is none). Then*

$$M(n) \ < \ 0.60\, n \log n + 52\, n \quad \textit{for all } n \geq 3, \qquad \textit{and} \qquad \limsup_{n\to\infty} \frac{M(n)}{n \log n} \ = \ \frac{1}{2}.$$

The limit superior is attained along $n = p\# + 2$, where $p\#$ is the product of the primes $\leq p$, by the sets $[n-1]_{r\mapsto mr}$ with $r = p\#$ and $m = \frac{q-1}{2}$, q the prime following p.

Corollary 1.5. *There is no constant C such that every tight set with n runners has all speeds at most Cn.*

Goddyn and Wong observed that, for a tight set $[n-1]_{r\mapsto mr}$, the speed r "(and thus n) must be quite large compared to integers m and s", and they studied the least r admissible for given (m, s) [18, Sec. 4]. Theorem 1.4 answers the inverse question: given the number of runners, how large can the accelerated speed be?

1.4 Further results and organisation

Section 2 describes the uncovered region. Its last part deals with the *deletion values* $\mathrm{LR}([n-1] \setminus \{r\})$, which bound $\mathrm{LR}([n-1]_{r\to w})$ from above. For $r \leq \frac{n-1}{2}$ the exact values are due to Liu:

the case $s = 1$ of his theorem [22, Thm. 18] gives $\mathrm{LR}([n-1] \setminus \{r\}) = 2/D_r$ with D_r the least integer $\geq n + r$ coprime to r; in particular

$$\mathrm{LR}\big([n-1] \setminus \{2\}\big) = \frac{2}{2\lfloor n/2 \rfloor + 3} \qquad (n \geq 5),$$

for which we include a short self-contained proof (Theorem 2.11). Beyond Liu's range, a single rational witness gives $\mathrm{LR}([n-1]\setminus\{r\}) > \frac{1}{n-1}$ whenever some integer in $[n+r, 2n-3]$ is coprime to r, and in particular for all $\frac{n-1}{2} < r \leq n-2-2\sqrt{n}$ (Theorem 2.9 and Corollary 2.10). Sections 3 and 4 prove Theorems 1.2 and 1.4.

Section 5 recalls the multi-acceleration theorem of Goddyn and Wong, which deserves to be better known, and extends the growth bound to the families it produces. Section 6 treats two-entry modifications. We show by a covering-density argument that a two-entry modification removing a speed q with $2 \leq q \leq n/10$ is never tight once $n \geq \max(40, 10q)$, irrespective of the other removed speed and of the inserted speeds (Theorem 6.6). An exact census covers $8 \leq n \leq 140$ with inserted speeds at most $8n$ (Theorem 6.10). Section 7 describes the computations, Section 8 collects open problems, and Appendix A comments on the shifted variant of the conjecture.

On the methods. The proofs of Theorems 1.2 and 1.4 use only Dirichlet's theorem, interval-covering arguments, Kanold's bound on the Jacobsthal function, and classical estimates for Chebyshev's function. Beyond these, they rely on Theorem 1.1 and on the tightness of T_1 and T_2 [34]. The only finite verification they need is the inequality $\theta(x) \geq 0.84\,x$ for $101 \leq x < 523$, which can be read off from a table of primes. The census of Section 6, the small cases of Theorem 6.4 and the spectrum search of Section 7.1 are computer-assisted: each is a proof by finite enumeration in exact rational arithmetic, and the programs and logs are publicly available.

2 The uncovered region

Throughout, log is the natural logarithm. For a finite set W we put $f_W(t) = \min_{v \in W} \|vt\|$, and for $w \in \mathbb{N}$ we write $B_w = \{t \in [0,1) : \|wt\| \leq \frac{1}{n}\}$ for the *bad set* of w. For $R \subseteq [n-1]$ let

$$U(n,R) = \Big\{ t \in [0,1) : \ \|it\| > \tfrac{1}{n} \text{ for all } i \in [n-1] \setminus R \Big\}, \qquad U(n,r) = U(n,\{r\}).$$

Since enlarging R removes constraints,

$$R \subseteq R' \quad \Longrightarrow \quad U(n,R) \subseteq U(n,R'). \tag{2}$$

2.1 Basic lemmas

Lemma 2.1 (Dirichlet)**.** *For every $t \in \mathbb{R}$ there is $i \in [n-1]$ with $\|it\| \leq \frac{1}{n}$. Consequently $U(n,\emptyset) = \emptyset$ and $\mathrm{LR}([n-1]) = \frac{1}{n}$.*

Proof. The first assertion is Dirichlet's theorem. For the second, take $t = a/n$ with $\gcd(a,n) = 1$; then $\|it\| \geq \frac{1}{n}$ for all $i \in [n-1]$. □

Lemma 2.2 (Coverage)**.** *Let $V = ([n-1]\setminus R) \cup W$ with $\mathrm{LR}(V) \leq \frac{1}{n}$. Then $U(n,R) \subseteq \bigcup_{w \in W} B_w$.*

Proof. For $t \in U(n,R)$ every retained speed i has $\|it\| > \frac{1}{n}$, while $f_V(t) \le \frac{1}{n}$; hence $\|wt\| \le \frac{1}{n}$ for some $w \in W$. □

Lemma 2.3 (Bad intervals)**.** *Let $n \ge 3$ and $w \in \mathbb{N}$. Then B_w is the disjoint union (on the circle) of the closed intervals $\left[\frac{j}{w} - \frac{1}{wn}, \frac{j}{w} + \frac{1}{wn}\right]$, $j \in \mathbb{Z}$, each of length $\frac{2}{wn}$. In particular, every connected subset of B_w lies in one of these intervals.*

Lemma 2.4 (Divisibility)**.** *If $2r > n-1$ and $[n-1]_{r\to w}$ is tight, then $r \mid w$.*

Proof. Since $2r > n-1$, no $i \in [n-1] \setminus \{r\}$ is divisible by r, so $\|i/r\| \ge \frac{1}{r} > \frac{1}{n}$ for these i, i.e. $\frac{1}{r} \in U(n,r)$. By Lemma 2.2, $\|w/r\| \le \frac{1}{n} < \frac{1}{r}$. As $\|w/r\| \in \frac{1}{r}\mathbb{Z}$, this forces $r \mid w$. □

2.2 The components of $U(n,r)$

For $r \ge 2$ and $x \not\equiv 0 \pmod r$ let

$$F(x) = \max\{i \le n-1 :\ i \ne r,\ i \equiv x \pmod r\}, \qquad I(n,r) = \min\{i \ge n-r :\ \gcd(i,r) = 1\}.$$

The block $\{s, s+1, \dots, n-1\}$, $s = n-r$, consists of r consecutive integers, so every residue class modulo r has exactly one representative in it. The only multiple of r that is excluded from F, namely r itself, lies in the zero class. Hence, for every unit x modulo r, $F(x)$ is the representative of x in $\{s, \dots, n-1\}$, and

$$\{F(x) :\ \gcd(x,r) = 1\} = \{i \in [s, n-1] :\ \gcd(i,r) = 1\}, \qquad I(n,r) = \min_{\gcd(x,r)=1} F(x). \tag{3}$$

For a reduced residue p modulo r we write $u_p \equiv p^{-1} \pmod r$.

Proposition 2.5 (Components, middle range)**.** *Let $2 \le r \le \frac{n-1}{2}$, so that $s = n - r \ge r + 1$. Then*

$$U(n,r) = \bigsqcup_{\substack{0 \le p < r \\ \gcd(p,r)=1}} (L_p \sqcup R_p), \qquad L_p = \left(\tfrac{p}{r} - \tfrac{s}{rnF(u_p)},\ \tfrac{p}{r} - \tfrac{1}{2rn}\right), \quad R_p = \left(\tfrac{p}{r} + \tfrac{1}{2rn},\ \tfrac{p}{r} + \tfrac{s}{rnF(r-u_p)}\right),$$

and all these intervals are nonempty.

Proof. Trapping. Let $t \in U(n,r)$. By Lemma 2.1 some $i \in [n-1]$ has $\|it\| \le \frac{1}{n}$; by definition of $U(n,r)$ this i must be r, so $|t - \frac{j}{r}| \le \frac{1}{rn}$ for some $j \in \mathbb{Z}$. If $g = \gcd(j,r) > 1$, then $r' = r/g \in [n-1] \setminus \{r\}$ satisfies $\|r't\| \le r'|t - \frac{j}{r}| \le \frac{1}{gn} < \frac{1}{n}$, which is impossible. Hence $j \equiv p \pmod r$ with $\gcd(p,r) = 1$, and we may write $t = \frac{p}{r} + \delta$ with $|\delta| \le \frac{1}{rn}$. It remains to determine which such δ satisfy $\|it\| > \frac{1}{n}$ for all $i \in [n-1] \setminus \{r\}$.

Multiples of r. For $i = kr$ with $2 \le k \le \frac{n-1}{r}$ we have $\|it\| = kr|\delta|$, because $kr|\delta| \le \frac{k}{n} < \frac{1}{2}$. The constraint reads $|\delta| > \frac{1}{krn}$, and since $2r \le n-1$ the strongest one is

$$|\delta| > \tfrac{1}{2rn}. \tag{4}$$

Non-multiples. Let $r \nmid i$, $\rho \equiv ip \pmod r$ with $1 \le \rho \le r-1$, and $d = \min(\rho, r-\rho)$. Then $it \equiv x := \frac{\rho}{r} + i\delta \pmod 1$, and $|i\delta| \le \frac{n-1}{rn} < \frac{1}{r}$, so $x \in (\frac{\rho-1}{r}, \frac{\rho+1}{r}) \subseteq [0,1]$. If $i\delta$ moves x towards the integer nearest to $\frac{\rho}{r}$ (for $\rho = \frac{r}{2}$ both directions count), then $\|it\| = \frac{d}{r} - |i\delta|$. Otherwise $\rho \ne \frac{r}{2}$, so $r \ge 3$ and $d \le \frac{r-1}{2}$, and $\|it\| \ge \min\left(\frac{d}{r}, \frac{r-d-1}{r}\right) \ge \frac{1}{r} > \frac{1}{n}$. So the only active constraint is $|\delta| < \frac{dn-r}{rni}$, on the side that moves x towards an integer.

If $d \geq 2$ this constraint is void on the trap, because $\frac{dn-r}{rni} \geq \frac{2n-r}{rn(n-1)} = \frac{n+s}{rn(n-1)} > \frac{1}{rn}$. If $d = 1$, then $i \equiv \pm u_p \pmod r$. The class $i \equiv u_p$ (so $\rho = 1$) constrains $\delta < 0$, the class $i \equiv -u_p$ (so $\rho = r-1$) constrains $\delta > 0$, and in each class the largest admissible i, namely $F(u_p)$ resp. $F(r-u_p)$, gives the strongest bound $|\delta| < \frac{s}{rnF(\cdot)}$. (For $r = 2$ the two classes coincide, and $F(1)$ bounds both sides.)

Together with (4) this yields exactly $L_p \sqcup R_p$. Both intervals are nonempty because $F \leq n-1 < 2s$. Conversely, every point of $L_p \sqcup R_p$ satisfies all the constraints above. □

Proposition 2.6 (Components, large range)**.** *Let $2 \leq r \leq n-1$ with $2r > n-1$, i.e. $r \geq s$. Then*

$$U(n,r) = \bigsqcup_{\substack{0 \leq p < r \\ \gcd(p,r)=1}} \Big(\tfrac{p}{r} - \tfrac{s}{rnF(u_p)},\ \tfrac{p}{r} + \tfrac{s}{rnF(r-u_p)} \Big),$$

where in this range $F(x) = x$ for $s \leq x < r$ and $F(x) = x + r$ for $1 \leq x < s$. Moreover $\sup_{t \in U(n,r)} \|rt\| = \frac{s}{n\, I(n,r)}$.

Proof. The trapping step and the analysis of non-multiples are exactly as in Proposition 2.5; they used only $F(x) \geq s$, which holds by (3). Since $2r > n-1$ there are no multiples $kr \leq n-1$ with $k \geq 2$, so (4) disappears and $\frac{p}{r} \in U(n,r)$. On the component around $\frac{p}{r}$ we have $\|rt\| = r|t - \frac{p}{r}|$, whose supremum is $\frac{s}{n} \min\big(F(u_p), F(r-u_p)\big)^{-1}$; maximising over p and using (3) gives $\frac{s}{nI(n,r)}$. □

Corollary 2.7 (Longest component)**.** *Let $\ell(n,r)$ be the length of the longest connected component of $U(n,r)$. Then*

$$\ell(n,r) = \frac{2s - I(n,r)}{2rn\, I(n,r)} \ \text{ if } 2r \leq n-1, \qquad \ell(n,r) = \frac{s}{rn} \max_{\gcd(u,r)=1} \Big(\frac{1}{F(u)} + \frac{1}{F(r-u)} \Big) \ \text{ if } 2r > n-1.$$

Proof. In the middle range $|R_p| = \frac{2s - F(r-u_p)}{2rnF(r-u_p)}$ and similarly for L_p; these lengths decrease in F, so by (3) the maximum is attained at $F = I(n,r)$. The large range is immediate from Proposition 2.6. □

For $r = 2$ the uncovered region is a pair of intervals around $\frac{1}{2}$, which we call the *shell*; it will be used both in Section 3 and in Section 6.

Corollary 2.8 (The shell)**.** *Let $n \geq 5$. Then $U(n,2) = L_- \sqcup L_+$ with*

$$L_- = \big(\tfrac{1}{2} - \beta,\ \tfrac{1}{2} - \tfrac{1}{4n}\big), \qquad L_+ = \big(\tfrac{1}{2} + \tfrac{1}{4n},\ \tfrac{1}{2} + \beta\big), \qquad \beta = \begin{cases} \frac{1}{2n} & n \text{ odd}, \\ \frac{n-2}{2n(n-1)} & n \text{ even}, \end{cases}$$

and both components have length $\Delta(n) = \beta - \frac{1}{4n}$, which equals $\frac{1}{4n}$ for odd n and $\frac{n-3}{4n(n-1)}$ for even n.

Proof. Apply Proposition 2.5 with $r = 2$ (the middle range $4 \leq n-1$): the only reduced centre is $\frac{1}{2}$, and $F(1)$ is the largest odd integer $\leq n-1$. □

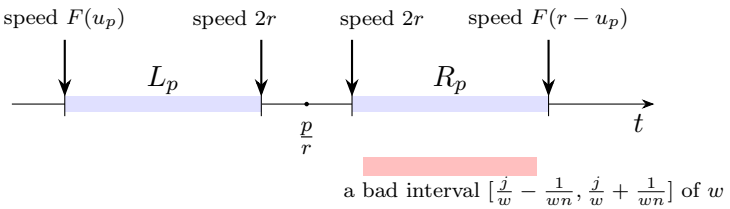


Figure 1: The two components of $U(n,r)$ next to p/r in the middle range $2r \le n-1$ (Proposition 2.5). The inner endpoints are determined by the speed $2r$, the outer ones by $F(u_p)$ and $F(r-u_p)$. If $[n-1]_{r\to w}$ is tight, each component lies in a single bad interval of w.

2.3 Deletion values

Adding a speed cannot increase LR, and $[n-1]\setminus\{r\}$ is contained in $[n-1]$. Therefore

$$\mathrm{LR}\big([n-1]_{r\to w}\big) \;\le\; \mathrm{LR}\big([n-1]\setminus\{r\}\big) \qquad \text{and} \qquad \mathrm{LR}\big([n-1]\setminus\{r\}\big) \;\ge\; \tfrac{1}{n}. \tag{5}$$

The Lonely Runner Conjecture for the $n-2$ speeds $[n-1]\setminus\{r\}$ predicts the stronger bound $\mathrm{LR}([n-1]\setminus\{r\}) \ge \frac{1}{n-1}$, with equality for $r=n-1$ by Lemma 2.1. For $r \le \frac{n-1}{2}$ the deletion values are in fact known exactly: Liu [22, Thm. 18] determined $\kappa([m]\setminus\{k,2k,\dots,sk\})$ for $m \ge (s+1)k$, and the case $s=1$ (with $m=n-1$, $k=r$) gives $\mathrm{LR}([n-1]\setminus\{r\}) = 2/D_r$, where D_r is the least integer $D \ge n+r$ with $\gcd(D,r)=1$. The following witness argument recovers the lower bound $2/D_r$ and extends it beyond $r \le \frac{n-1}{2}$.

Theorem 2.9 (Rational witnesses)**.** *Let $n \ge 3$, $r \in [n-1]$, and let $D \ge n+r$ be an integer with $\gcd(D,r)=1$. Then*

$$\mathrm{LR}\big([n-1]\setminus\{r\}\big) \;\ge\; \frac{2}{D}.$$

Consequently $\mathrm{LR}([n-1]\setminus\{r\}) > \frac{1}{n-1}$ if some integer in $[n+r,\,2n-3]$ is coprime to r, and $\mathrm{LR}([n-1]\setminus\{r\}) \ge \frac{1}{n-1}$ if some integer in $[n+r,\,2n-2]$ is coprime to r.

Proof. Choose a with $ar \equiv 1 \pmod D$ and put $t = a/D$; then $\gcd(a,D)=1$. Let $i \in [n-1]\setminus\{r\}$. Since $0 < i < D$, we have $ia \not\equiv 0 \pmod D$. If $ia \equiv \pm 1 \pmod D$, then $i \equiv \pm r \pmod D$, so $i = r$ or $i \ge D-r \ge n$; both are excluded. Hence $ia \bmod D \in [2, D-2]$, i.e. $\|it\| \ge \frac{2}{D}$. The consequences follow because $\frac{2}{D} > \frac{1}{n-1}$ for $D \le 2n-3$ and $\frac{2}{D} \ge \frac{1}{n-1}$ for $D \le 2n-2$. □

Recall that the *Jacobsthal function* $j(r)$ is the least k such that every k consecutive integers contain one coprime to r. Kanold proved $j(r) \le 2^{\omega(r)}$, where $\omega(r)$ is the number of distinct prime factors of r [20].

Corollary 2.10. *If $n-r-2 \ge j(r)$, then $\mathrm{LR}([n-1]\setminus\{r\}) > \frac{1}{n-1}$. In particular this holds for every $r \le n-2-2\sqrt{n}$.*

Proof. The interval $[n+r, 2n-3]$ contains $n-r-2$ consecutive integers. For the second claim note that $j(r) \le 2^{\omega(r)} \le \tau(r) \le 2\sqrt{r} < 2\sqrt{n}$, where $\tau(r)$ is the number of divisors of r. □

For $r=2$ the witness of Theorem 2.9 is optimal; the result is the case $r=2$ of Liu's theorem [22, Thm. 18], and we include a short self-contained proof.

Theorem 2.11 (Deletion value for $r = 2$; Liu)**.** *For every $n \geq 5$,*

$$\mathrm{LR}\big([n-1] \setminus \{2\}\big) = \frac{2}{D}, \qquad D = 2\lfloor n/2 \rfloor + 3.$$

Proof. Put $V = [n-1] \setminus \{2\}$. The integer D is odd and equals $n+2$ or $n+3$, so Theorem 2.9 gives $\mathrm{LR}(V) \geq \frac{2}{D}$.

For the upper bound let $t \in [0,1)$. Among the $n-1$ points $0, t, \ldots, (n-2)t$ of the circle two are at distance at most $\frac{1}{n-1}$, so some $d \in [n-2]$ has $\|dt\| \leq \frac{1}{n-1} \leq \frac{2}{D}$; the last inequality is $D \leq 2n-2$, valid for $n \geq 5$. If $d \neq 2$ we are done. Suppose $d = 2$, and write $2t = m + \epsilon$ with $m \in \mathbb{Z}$ and $|\epsilon| = e = \|2t\| \leq \frac{1}{n-1}$.

- If $e \leq \frac{1}{D}$, then $\|4t\| = 2e \leq \frac{2}{D}$ (here $4 \in V$, and $2e \leq \frac{1}{2}$).
- If $e > \frac{1}{D}$ and m is even, then $\|t\| = \frac{e}{2} \leq \frac{1}{2(n-1)} < \frac{2}{D}$.
- If $e > \frac{1}{D}$ and m is odd, let i_0 be the largest odd element of V, namely $n-1$ or $n-2$; in both cases $i_0 = D-4 \geq 3$. Then $i_0 t \equiv \frac{1}{2} + \frac{i_0 \epsilon}{2} \pmod 1$ and $\frac{i_0 e}{2} \leq \frac{1}{2}$, so

$$\|i_0 t\| = \frac{1}{2} - \frac{i_0 e}{2} < \frac{1}{2} - \frac{D-4}{2D} = \frac{2}{D}.$$

In every case $f_V(t) \leq \frac{2}{D}$. □

Remark 2.12. The witness bound is not sharp in general: for $(n, r) = (5, 3)$ it gives $\frac{1}{4}$, whereas $t = \frac{1}{3}$ shows $\mathrm{LR}(\{1, 2, 4\}) \geq \frac{1}{3}$. Note also that $\frac{2}{D} > \frac{1}{n-1}$ in Theorem 2.11, so for $r = 2$ the interval in (5) contains the whole range $(\frac{1}{n}, \frac{1}{n-1})$.

3 Proof of Theorem 1.2

3.1 The middle range $3 \leq r \leq \frac{n-1}{2}$

Let $3 \leq r \leq \frac{n-1}{2}$, $s = n - r \geq r + 1$, $w \geq n$, and suppose that $V = [n-1]_{r \to w}$ is tight. Write

$$F_+ = F(1), \qquad F_- = F(r-1), \qquad w \equiv c \pmod r, \quad c \in \left(-\tfrac{r}{2}, \tfrac{r}{2}\right].$$

By Lemma 2.2, Proposition 2.5 (with $p = 1$) and continuity, $\|wt\| \leq \frac{1}{n}$ on the closures $\overline{L_1}$ and $\overline{R_1}$ of the two components next to $t_0 = \frac{1}{r}$.

Lemma 3.1 (Windows)**.** *Assume moreover $w < r(n-2)$. Then*

(R) $\|wt\| \leq \frac{1}{n}$ *on* $\overline{R_1}$ *implies* $c \leq \frac{r}{n} - \frac{ws}{nF_-}$*;*

(L) $\|wt\| \leq \frac{1}{n}$ *on* $\overline{L_1}$ *implies* $c \geq \frac{ws}{nF_+} - \frac{r}{n}$*.*

Proof. We prove (R); (L) is symmetric. Let $\overline{R_1} = [a, b]$ with $a = t_0 + \frac{1}{2rn}$ and $b = t_0 + \frac{s}{rnF_-}$. By Lemma 2.3 there is $j \in \mathbb{Z}$ with $wb - \frac{1}{n} \leq j \leq wa + \frac{1}{n}$. Write $w = Kr + c$, so that $wt_0 = K + \frac{c}{r}$, and put $J = j - K$. Then

$$\frac{c}{r} + \frac{ws}{rnF_-} - \frac{1}{n} \;\leq\; J \;\leq\; \frac{c}{r} + \frac{w}{2rn} + \frac{1}{n}.$$

If $J \geq 1$, the right inequality gives $c \geq r - \frac{w}{2n} - \frac{r}{n} > \frac{r}{2}$, where the last step is equivalent to $w < r(n-2)$; this contradicts $c \leq \frac{r}{2}$. If $J \leq -1$, the left inequality gives $c \leq -r + \frac{r}{n} - \frac{ws}{nF_-} < -\frac{r}{2}$, again impossible. Hence $J = 0$, and the left inequality is the claim. □

Lemma 3.2 (Signs)**.** *If $s \geq r+1$ and $w \geq n$, then $\frac{r}{n} - \frac{ws}{nF_-} < 0$ and $\frac{ws}{nF_+} - \frac{r}{n} > 0$.*

Proof. Both inequalities mean $ws > rF_\pm$. Since $F_\pm \leq n-1$ and $w \geq n$, it suffices that $ns > r(n-1)$, i.e. $n(s-r)+r > 0$. □

Lemma 3.3 (Key inequality)**.** *Let $r \geq 3$, $n \geq 2r+1$, $s = n-r$ and $I = I(n,r)$. Then $I < 2s$ and*

$$\frac{4rI}{2s-I} < r(n-2).$$

Proof. Put $d = I - s \geq 0$ and $f(n) = \frac{s(n-6)}{n+2}$, which is positive since $n \geq 7$. A short computation shows that the assertion is equivalent to $d < f(n)$, which also gives $I < s + s\frac{n-6}{n+2} < 2s$. The derivative of f (with r fixed) has the sign of $n^2+4n-8r-12$, which is at least $4r^2+4r-7 > 0$ for $n \geq 2r+1$; so f is increasing on $[2r+1, \infty)$. The d consecutive integers $s, \ldots, s+d-1$ all have a common factor with r, hence

$$d \;\leq\; j(r) - 1 \;\leq\; 2^{\omega(r)} - 1.$$

Case $\omega(r) \geq 2$. Then $r \geq 6$, and r is at least the product of $\omega(r)$ distinct primes, so $r \geq 2 \cdot 3^{\omega(r)-1}$. Since $2^k - 1 \leq 3^{k-1}$ for all $k \geq 1$, we get $d \leq \frac{r}{2}$. The inequality $\frac{r}{2} < f(n)$ is equivalent to $2n^2 - 3(r+4)n + 10r > 0$. At $n = 2r+1$ the left side equals $2r^2 - 9r - 10 > 0$ (as $r \geq 6$), and its derivative $4n - 3(r+4)$ is positive for $n \geq 2r+1$. Hence $d \leq \frac{r}{2} < f(n)$.

Case $r = p^a$ a prime power. Then $j(r) = 2$, so $d \leq 1$, and $d = 0$ unless $p \mid s$. If $r \geq 4$, then $f(n) \geq f(2r+1) = \frac{(r+1)(2r-5)}{2r+3} > 1$, because $2r^2 - 5r - 8 > 0$. If $r = 3$ and $3 \mid s$, then $s \geq 6$, so $n \geq 9$ and $f(n) \geq f(9) = \frac{18}{11} > 1$. In all cases $d < f(n)$. □

Proposition 3.4. *For $3 \leq r \leq \frac{n-1}{2}$ and $w \geq n$, the set $[n-1]_{r\to w}$ is not tight.*

Proof. Suppose it is tight. By Lemma 2.2 the longest component of $U(n,r)$ lies in one bad interval of w. Comparing lengths with Corollary 2.7 and Lemma 2.3 gives $\frac{2s-I}{2rnI} \leq \frac{2}{wn}$, i.e. $w \leq \frac{4rI}{2s-I}$, and hence $w < r(n-2)$ by Lemma 3.3. Now Lemmas 3.1 and 3.2 give $c < 0$ from (R) and $c > 0$ from (L), a contradiction. □

3.2 The boundary case $r = 2$

Proposition 3.5. *Let $n \geq 5$ and $w \geq n$. Then $[n-1]_{2\to w}$ is tight if and only if $(n,w) \in \{(5,7),(6,9)\}$.*

Proof. The sets T_1 and T_2 are tight by [34]. Conversely, let $V = [n-1]_{2\to w}$ be tight, and let $L_\pm$, β, $\Delta = \Delta(n)$ be as in Corollary 2.8. By Lemmas 2.2 and 2.3, $\overline{L_-}$ lies in a single bad interval $[c-h, c+h]$ with $c = \frac{j}{w}$ and $h = \frac{1}{wn}$, that is,

$$c - h \leq \tfrac{1}{2} - \beta, \qquad c + h \geq \tfrac{1}{2} - \tfrac{1}{4n}, \qquad 2h \geq \Delta. \tag{6}$$

The centre lies below $\frac{1}{2}$. If $c \geq \frac{1}{2}$, then $c - h \geq \frac{1}{2} - \frac{1}{wn} > \frac{1}{2} - \frac{1}{4n}$ because $w > 4$, so the interval misses the right endpoint of $\overline{L_-}$.

Parity. The largest centre $\frac{j}{w}$ below $\frac{1}{2}$ is $\frac{1}{2} - \frac{1}{w}$ if w is even and $\frac{1}{2} - \frac{1}{2w}$ if w is odd. Inserting it into the middle condition of (6) gives $w \geq 4(n-1)$ for even w and $w \geq 2(n-2)$ for odd w.

Length. The last condition of (6) reads $w \leq \frac{2}{n\Delta}$, i.e. $w \leq 8$ for odd n and $w \leq \frac{8(n-1)}{n-3}$ for even n.

For even w the bounds $4(n-1) \le w \le \frac{2}{n\Delta}$ force $n \le 3$ for odd n and $n \le 5$ for even n, contradicting $n \ge 5$. So w is odd and $2(n-2) \le w \le \frac{2}{n\Delta}$. For odd n this gives $n = 5$ and $w = 7$. For even n it gives $(n-2)(n-3) \le 4(n-1)$, i.e. $n^2 - 9n + 10 \le 0$, whence $n = 6$ and $w \in \{9, 11, 13\}$.

It remains to exclude $(n, w) = (6, 11)$ and $(6, 13)$. The point $t^* = \frac{87}{200}$ lies in $L_- = \left(\frac{13}{30}, \frac{11}{24}\right)$, and

$$\|t^*\| = \tfrac{87}{200}, \quad \|3t^*\| = \tfrac{61}{200}, \quad \|4t^*\| = \tfrac{52}{200}, \quad \|5t^*\| = \tfrac{35}{200}, \quad \|11t^*\| = \tfrac{43}{200}, \quad \|13t^*\| = \tfrac{69}{200},$$

all of which exceed $\frac{1}{6}$. Hence $\mathrm{LR}(V) > \frac{1}{6}$ for $w \in \{11, 13\}$. □

Remark 3.6. The window argument of Lemma 3.1 requires $w < r(n-2)$, whereas for $r = 2$ the covering condition forces $w \ge 2(n-2) = r(n-2)$. The two arguments meet exactly at the boundary, which explains why $r = 2$ is the only removed speed in the middle range that admits tight modifications.

3.3 Completion of the proof

Proof of Theorem 1.2. $2r > n-1$. By Lemma 2.4, $r \mid w$, and Theorem 1.1 applies; the exceptions $n = 2$ and $(3, 1, 4)$ do not occur for $n \ge 4$. This case contains $r = 2$ for $n = 4$.

$r = 1$. Here $w = 1 \cdot w$ is of Goddyn–Wong form, and (1) fails because $\gcd(1, b) = 1$; so the set is not tight, in accordance with (i).

$r = 2$, $n \ge 5$. This is Proposition 3.5; condition (i) fails by Remark 1.3(a).

$3 \le r \le \frac{n-1}{2}$. This is Proposition 3.4; again (i) fails by Remark 1.3(a). □

4 Proof of Theorem 1.4

Let $\theta(x) = \sum_{p \le x} \log p$ denote Chebyshev's function. By Theorem 1.2, for $n \ge 4$ a tight set $[n-1]_{r \to w}$ is either T_1 or T_2 (so $w \le 9$), or $w = mr$ with $m \ge 2$ and (1). In the latter case $s \ge 2$: for $s = 1$ the integer $b = 1$ would violate (1). Moreover every prime b in $[s, ms - 1]$ satisfies $\gcd(r, b) > 1$, i.e. $b \mid r$. As these primes are distinct, their product is at most $r < n$, so

$$\theta(ms - 1) - \theta(s - 1) \;\le\; \log r \;<\; \log n. \tag{7}$$

For $n = 3$ the only tight one-entry modification is $[2]_{1 \mapsto 4}$, so $M(3) = 4$.

Proof of the upper bound. We use the classical estimates of Rosser and Schoenfeld [29], $\theta(x) < 1.01624\, x$ for all $x > 0$ and $\theta(x) > x\left(1 - \frac{1}{\log x}\right)$ for $x \ge 41$, whence in particular

$$\theta(x) \;\ge\; 0.84\, x \qquad (x \ge 101)\,: \tag{8}$$

for $x \ge 523$ this follows from $1 - \frac{1}{\log x} > 0.84$, and for $101 \le x < 523$ it is checked directly from the list of primes below 523.

Let $n \ge 4$ and consider a tight $[n-1]_{r \mapsto mr}$ of Goddyn–Wong type. If $ms - 1 \ge 101$, then (7) and the two estimates give

$$0.84\,(ms - 1) \;<\; \log n + 1.01624\,(s - 1), \qquad \text{hence} \qquad ms \;<\; 1.1905 \log n + 1.2099\,(s - 1) + 1.$$

Dividing by $s \geq 2$ we obtain

$$m \;<\; 1.2099 + \frac{1.1905 \log n - 0.2099}{s} \;\leq\; 0.5953 \log n + 1.105,$$

and therefore $w = mr < mn < 0.60\, n \log n + 1.11\, n$. If $ms - 1 < 101$, then $m \leq \frac{101}{s} \leq 50.5$ and $w < mn \leq 50.5\, n$. Together with the sporadic speeds $w \leq 9$ this gives $M(n) < \max(0.60\, n \log n + 1.11\, n,\ 50.5\, n,\ 9) < 0.60\, n \log n + 52\, n$ for $n \geq 4$, and the bound is trivial for $n = 3$. □

Proof that $\limsup M(n)/(n \log n) \leq \frac{1}{2}$. Fix $\varepsilon \in (0, \frac{1}{2})$. By the prime number theorem there is x_ε with $\theta(x) \geq (1-\varepsilon)x$ for $x \geq x_\varepsilon$. Consider a tight $[n-1]_{r \mapsto mr}$ as above. If $ms - 1 < x_\varepsilon$, then $w < mn \leq x_\varepsilon n$. Otherwise (7) gives $(1-\varepsilon)(ms-1) < \log n + 1.02\, s$, so that, using $s \geq 2$,

$$m \;<\; \frac{\log n}{2(1-\varepsilon)} + \frac{1.02}{1-\varepsilon} + \frac{1}{2} \;<\; \frac{\log n}{2(1-\varepsilon)} + 3.$$

Hence $M(n) \leq \frac{n \log n}{2(1-\varepsilon)} + \max(3, x_\varepsilon, 9)\, n$ for $n \geq 4$, and letting $\varepsilon \to 0$ proves the claim. □

Proof of the lower bound. Let $p \geq 3$ be prime, q the next prime, $r = p\#$, $n = r + 2$ (so $s = 2$), and $m = \frac{q-1}{2} \geq 2$. Every integer $b \in [2, 2m-1] = [2, q-2]$ has a prime factor at most $q - 2$, hence at most p, which divides r. Thus (1) holds, and $[n-1]_{r \mapsto mr}$ is tight by Theorem 1.1, with

$$w = mr = \frac{q-1}{2}\,(n-2).$$

Since $\log n = \theta(p) + o(1)$ and $\theta(p) \sim p \sim q$ by the prime number theorem, $w = (\frac{1}{2} + o(1))\, n \log n$. The multiplier m is optimal for this pair (r, s): for $m+1$ the interval $[2, 2m+1]$ contains the prime $q \nmid r$. □

Remark 4.1. Along the primorials the ratio $w/(n \log n)$ approaches $\frac{1}{2}$ slowly and not monotonically, since it depends on the gap $q - p$. Its largest value is 0.9246, at $p = 7$ ($n = 212$, $r = 210$, $w = 1050$), and it still exceeds 0.67 at $p = 37$, where $n \approx 7.4 \cdot 10^{12}$. In the same vein, the term $52\, n$ in Theorem 1.4 dominates $0.60\, n \log n$ for all n below about $4 \cdot 10^{37}$; at computable sizes the effective content of the upper bound is a linear bound, and the constant $\frac{1}{2}$ is a genuinely asymptotic statement.

5 Several accelerated runners

The sufficiency part of Theorem 1.1 is a special case of a more general result of Goddyn and Wong, in which several runners are accelerated simultaneously.

Theorem 5.1 (Goddyn–Wong [18, Thm. 3.1])**.** *Let $m_1, \ldots, m_{n-1}$ be positive integers and $[n-1]_{\mathbf{m}} := \{m_1, 2m_2, \ldots, (n-1)m_{n-1}\}$. If*

$$\gcd(r, b) > 1 \quad \textit{for every } r \in [n-1] \textit{ with } m_r \geq 2 \textit{ and every } b \in [\, n - r,\ m_r(n-r) - 1\,], \tag{9}$$

then $\mathrm{LR}([n-1]_{\mathbf{m}}) = \frac{1}{n}$.

Goddyn and Wong illustrate Theorem 5.1 by the tight set $([73] \setminus \{70, 72\}) \cup \{140, 144\}$, $n = 74$, which is also discussed by Tao [31], and they construct tight sets in which arbitrarily many runners are accelerated [18, Sec. 3]. The following simple pattern with all multipliers equal to 2 makes the case of up to three accelerated runners explicit.

Example 5.2 (Doublings)**.** For $1 \le k \le 3$ let $s_i = 2^i$ and let P_i be the product of the primes in $[s_i, 2s_i - 1]$, so $P_1 = 6$, $P_2 = 35$, $P_3 = 143$. If

$$n \equiv s_i \pmod{P_i} \qquad (1 \le i \le k),$$

then $V = \big([n-1] \setminus \{n - s_i\}_{i \le k}\big) \cup \{2(n - s_i)\}_{i \le k}$ is tight. By the Chinese Remainder Theorem the admissible n form the residue classes $n \equiv 2 \pmod 6$ ($k = 1$; take $n \ge 8$), $n \equiv 74 \pmod{210}$ ($k = 2$) and $n \equiv 23174 \pmod{30030}$ ($k = 3$).

Indeed, put $r_i = n - s_i$ and let $b \in [s_i, 2s_i - 1]$. If b is prime, then $b \mid P_i \mid r_i$. If b is composite, then, as $i \le 3$, b is divisible by 2 or by 3. All r_i are even because $n \equiv 2 \pmod 6$. The only odd composites that occur are 9 and 15 (for $i = 3$), and $3 \mid r_3$ since $r_3 = n - 8 \equiv 0 \pmod 3$. Hence (9) holds. The pattern breaks down at $k = 4$: the conditions force $r_4 \equiv 1 \pmod 3$ and $r_4 \equiv 2 \pmod 7$, so $\gcd(r_4, 21) = 1$ with $21 \in [16, 31]$.

The first instances with $k = 2$ are $n = 74$, the example of Goddyn and Wong, and $n = 284$. Both were also verified independently in exact arithmetic (Section 7).

The divisibility argument of Lemma 2.4 has the following variant, valid for arbitrary sets.

Lemma 5.3. *If $V \subset \mathbb{N}$ satisfies $\mathrm{LR}(V) \le \frac{1}{n}$, then for every $r \in [n-1]$ the set V contains a multiple of r.*

Proof. At $t = \frac{1}{r}$ some $v \in V$ has $\|v/r\| \le \frac{1}{n} < \frac{1}{r}$, which forces $r \mid v$. □

Corollary 5.4 (Growth bound under the Goddyn–Wong criterion)**.** *Let $V = ([n-1] \setminus R) \cup W$ be tight, and suppose that every $w \in W$ has the form $w = mr$ with $r \in R$, $m \ge 2$, and (1) for the pair (r, m). Then $\max V < 0.60\, n \log n + 52\, n$. This applies to every set produced by Theorem 5.1.*

Proof. The proof of the upper bound in Theorem 1.4 used only (1) for the pair (r, m), and applies to each $w \in W$ separately. □

6 Two-entry modifications

We now consider $V = ([n-1] \setminus R) \cup W$ with $R = \{r_1, r_2\}$ and $W = \{w_1, w_2\}$, $n \le w_1 < w_2$. By Lemma 2.2, if V is tight then

$$U(n, R) \;\subseteq\; B_{w_1} \cup B_{w_2}.$$

This is a *union* condition. Different parts of one component may be covered by different speeds, so the window argument of Section 3 does not carry over directly. A counting argument does, however, as long as one removed speed is small.

6.1 Two tools

Lemma 6.1 (Covering density)**.** *Let $w \ge n$. For every interval $J \subseteq [0, 1)$,*

$$|B_w \cap J| \;\le\; \Big(w|J| + \tfrac{2}{n} + 1\Big)\frac{2}{wn}.$$

Proof. A bad interval meeting J has its centre $\frac{j}{w}$ in the $\frac{1}{wn}$-neighbourhood of J. So j ranges over an interval of length $w|J| + \frac{2}{n}$, which contains at most $w|J| + \frac{2}{n} + 1$ integers, and each bad interval has length $\frac{2}{wn}$. □

Lemma 6.2 (Edge reach)**.** *Let $q \geq 2$ and $\mu \geq 2$ be integers with $\mu q < n \leq w$, and put $a = \frac{1}{q} + \frac{1}{\mu q n}$. If some bad interval of w contains a, then $w \geq \mu(n-q)$.*

Proof. Let $c = \frac{j}{w}$ be the centre, so $|c-a| \leq \frac{1}{wn}$. Since $w > \mu q$ we have $\frac{1}{wn} < \frac{1}{\mu q n}$, hence $c > \frac{1}{q}$, so $jq - w$ is a positive integer and $\frac{1}{wq} \leq c - \frac{1}{q} \leq \frac{1}{\mu q n} + \frac{1}{wn}$. Multiplying by wqn gives $n \leq \frac{w}{\mu} + q$. □

6.2 Removing the speed 2

Theorem 6.3. *Let $r' \in [n-1] \setminus \{2\}$ be arbitrary. If $n \geq 19$ is odd or $n \geq 20$ is even, then no set $([n-1] \setminus \{2, r'\}) \cup \{w_1, w_2\}$ with $w_1, w_2 \geq n$ is tight.*

Proof. Suppose such a set is tight. By (2) and Corollary 2.8, $L_- \cup L_+ = U(n,2) \subseteq U(n, \{2, r'\})$, so $B_{w_1} \cup B_{w_2} \supseteq L_- \cup L_+$. As $B_{w_1} \cup B_{w_2}$ is closed, it contains $\inf L_+ = \frac{1}{2} + \frac{1}{4n}$, and Lemma 6.2 (with $q = \mu = 2$) shows that one of the speeds, say w_*, satisfies $w_* \geq 2(n-2)$; the other is at least n. Let $\ell = \Delta(n)$ be the common length of $L_\pm$. By Lemma 6.1, the part of B_w in each component has measure at most $\beta_w = (w\ell + \frac{2}{n} + 1)\frac{2}{wn}$, which decreases in w. Hence the measure that can be covered is at most

$$\Phi(n) = 2\big(\beta_{2(n-2)} + \beta_n\big),$$

and coverage is impossible if $\Phi(n) < 2\ell$. For odd n ($\ell = \frac{1}{4n}$) one finds $\Phi(n) = \frac{3n+2}{n^2(n-2)} + \frac{5n+8}{n^3}$, and $\Phi(n) < 2\ell$ is equivalent to $n^3 - 18n^2 + 32 > 0$, which holds for $n \geq 19$. For even n ($\ell = \frac{n-3}{4n(n-1)}$) the inequality $\Phi(n) < 2\ell$ is equivalent to

$$n^4 - 21n^3 + 30n^2 + 16n - 32 > 0,$$

which holds for $n \geq 20$: the left side is 4288 at $n = 20$ and increasing for $n \geq 15$. □

Theorem 6.4 ($R = \{2, 3\}$)**.** *Let $n \geq 8$ and $w_1, w_2 \geq n$. If $n \geq 19$, the set $([n-1] \setminus \{2,3\}) \cup \{w_1, w_2\}$ is never tight. If $8 \leq n \leq 18$ and $w_1, w_2 \leq 40n$, it is tight if and only if $n = 8$ and $\{w_1, w_2\} = \{11, 13\}$, i.e. $V = \{1, 4, 5, 6, 7, 11, 13\}$.*

Proof. The case $n \geq 19$ is Theorem 6.3. The tightness of $\{1, 4, 5, 6, 7, 11, 13\}$ is classical [18] and was confirmed in exact arithmetic. For $8 \leq n \leq 18$ the statement is a finite computation. All pairs $n \leq w_1 < w_2 \leq 40n$ surviving the edge-reach conditions are tested in exact rational arithmetic against the exact components of $U(n, \{2,3\})$ (Section 7). □

6.3 Arbitrary small removed speeds

We now run the argument of Theorem 6.3 at $\frac{1}{q}$ instead of $\frac{1}{2}$. Throughout this subsection

$$q \geq 2, \qquad 4q \leq n-1, \qquad R = \{q, r'\} \text{ with } r' \in [n-1] \setminus \{q\}, \qquad S = [n-1] \setminus R,$$

and

$$F_{\pm 1} = \max\{i \in S : \ i \equiv \pm 1 \pmod q\}, \qquad m_0 q = \min\{i \in S : \ q \mid i\}.$$

Each residue class has a representative in $\{n-q, \ldots, n-1\}$ and one in $\{n-2q, \ldots, n-q-1\}$, and at most one of them is removed; hence $n - 2q \leq F_{\pm 1} \leq n-1$. Likewise $m_0 \in \{2, 3\}$, since at most one of $2q, 3q \leq n-1$ is removed. From $4q \leq n-1$ we get

$$F_{\pm 1} \ \geq \ n - 2q \ \geq \ \frac{n+1}{2}, \qquad \beta_\pm := \frac{1/q - 1/n}{F_{\mp 1}} \ \leq \ \frac{n-q}{qn(n-2q)} \ < \ \frac{2}{q(n+1)} \ < \ \frac{1}{n}. \tag{10}$$

Lemma 6.5 (Windows at $1/q$)**.** *The intervals*

$$W_- = \Big(\tfrac{1}{q} - \beta_-,\ \tfrac{1}{q} - \tfrac{1}{m_0 qn}\Big), \qquad W_+ = \Big(\tfrac{1}{q} + \tfrac{1}{m_0 qn},\ \tfrac{1}{q} + \beta_+\Big)$$

are nonempty and contained in $U(n, R)$.

Proof. We treat W_+; for W_- replace τ by $-\tau$ and interchange the classes 1 and -1. Write $t = \frac{1}{q} + \tau$ with $\frac{1}{m_0 qn} < \tau < \beta_+$.

Nonemptiness. The inequality $\frac{1}{m_0 qn} < \beta_+$ is equivalent to $F_{-1} < m_0(n-q)$, which holds since $F_{-1} \le n - 1 < 2(n-q)$.

Multiples of q. Let $kq \in S$, so $k \ge m_0$. The constraint $\|kq\tau\| > \frac{1}{n}$ fails only for τ within $\frac{1}{kqn}$ of some $\frac{j}{kq}$. For $j = 0$ this region lies below $\frac{1}{m_0 qn}$. For $j \ge 1$ it lies above $\frac{1-1/n}{kq} \ge \frac{1}{n} > \beta_+$, because $kq \le n - 1$.

Other classes. Let $i \in S$ with $\rho = i \bmod q \in [1, q-1]$; then $it \equiv \frac{\rho}{q} + i\tau \pmod 1$. We claim $i\tau < \frac{q-\rho}{q} - \frac{1}{n}$. For $\rho = q - 1$ this follows from $i \le F_{-1}$. For $\rho \le q - 2$ we use $i \le n - 1 < 2F_{-1}$ from (10):

$$i\tau < \frac{i(n-q)}{qnF_{-1}} < \frac{2(n-q)}{qn} < \frac{2n-q}{qn} \le \frac{(q-\rho)n - q}{qn} = \frac{q-\rho}{q} - \frac{1}{n}.$$

Thus $z = \frac{\rho}{q} + i\tau \in (\frac{\rho}{q}, 1 - \frac{1}{n})$, and $\|z\| > \frac{1}{n}$: if $z \le \frac{1}{2}$, then $\|z\| \ge \frac{\rho}{q} \ge \frac{1}{q} > \frac{1}{n}$, and otherwise $\|z\| = 1 - z > \frac{1}{n}$. □

Theorem 6.6 (A small removed speed)**.** *Let* $q \ge 2$ *and* $n \ge \max(40, 10q)$*, let* $r' \in [n-1] \setminus \{q\}$ *be arbitrary, and let* $w_1, w_2 \ge n$*. Then* $([n-1] \setminus \{q, r'\}) \cup \{w_1, w_2\}$ *is not tight.*

Proof. The hypotheses imply $4q \le n - 1$. Suppose that the set is tight. Then $B_{w_1} \cup B_{w_2}$ covers $W_- \cup W_+$ by Lemmas 2.2 and 6.5, and, being closed, it contains the inner endpoint $\frac{1}{q} + \frac{1}{m_0 qn}$ of W_+. By Lemma 6.2 with $\mu = m_0$, one speed w^* satisfies $w^* \ge m_0(n-q) \ge 2(n-q)$; the other speed w' is at least n. Let $\Lambda = |W_-| + |W_+|$. Using $F_{\pm 1} \le n - 1$ and $m_0 \ge 2$,

$$\Lambda \ \ge\ 2\ell^*, \qquad \ell^* = \frac{n-q}{qn(n-1)} - \frac{1}{2qn} = \frac{n - 2q + 1}{2qn(n-1)}.$$

Applying Lemma 6.1 to both speeds and both windows,

$$\Lambda \ \le\ \frac{4\Lambda}{n} + \frac{4(1+2/n)}{n}\Big(\frac{1}{2(n-q)} + \frac{1}{n}\Big),$$

so coverage forces

$$\ell^*\Big(1 - \frac{4}{n}\Big) \ \le\ \frac{1 + 2/n}{n(n-q)} + \frac{2(1+2/n)}{n^2}. \tag{11}$$

We show that (11) fails. Using $n - 2q + 1 \ge \frac{4n}{5}$, $\frac{n-4}{n} \ge \frac{9}{10}$ and $n - 1 < n$,

$$\ell^*\Big(1 - \frac{4}{n}\Big) = \frac{(n-2q+1)(n-4)}{2qn^2(n-1)} \ \ge\ \frac{9}{25\,qn}.$$

Using $1 + \frac{2}{n} \le \frac{21}{20}$ and $n - q \ge \frac{9n}{10}$, the right side of (11) is at most $\frac{21}{20}\big(\frac{10}{9} + 2\big)\frac{1}{n^2} = \frac{49}{15n^2}$. Finally $\frac{9}{25qn} > \frac{49}{15n^2}$ is equivalent to $135n > 1225q$, which follows from $q \le \frac{n}{10}$. □

Remark 6.7 (Why density beats windows here)**.** Each speed's bad set has density exactly $2/n$ on large scales (w intervals of length $\frac{2}{wn}$), so two speeds cover at most a $\frac{4}{n}$-fraction of any region, up to the per-component slack of Lemma 6.1. Since $|L_- \cup L_+|$ is a fixed $\approx \frac{1}{2n}$ and no constraint of the second removed speed can shrink it ($U(n,2) \subseteq U(n,R)$ by (2), as used in the proof of Theorem 6.3), coverage is impossible once $\frac{4}{n}$ plus slack drops below 1. The argument is insensitive to the split phenomenon—it does not care *which* speed covers *which* part, within or across components—and requires no upper bound on the speeds. Run at $\frac{1}{q}$ rather than $\frac{1}{2}$, the mechanism needs a guaranteed window of measure $\gg \frac{1}{n^2}$ adjacent to $\frac{1}{q}$, which the class structure modulo q supplies as long as $q \le n/10$. What it cannot handle is two removals both close to $\frac{n}{2}$: the guaranteed region $U(n,r_1) \cup U(n,r_2)$ can have measure as small as $O\big(\frac{n-2r}{n^2}\big)$, which tends to 0 as $r \to \frac{n-1}{2}$, while the slack accumulates over its $\asymp \varphi(r)$ components. That regime is settled for $8 \le n \le 140$, $w \le 8n$ by the census (Theorem 6.10) and remains open in general (Question 8.1).

Remark 6.8 (Sharper thresholds)**.** After clearing denominators, the failure of (11) is equivalent to

$$G(n,q) = n(n-q)(n-4)(n-2q+1) - 2q(n-1)(n+2)(3n-2q) > 0,$$

whose leading part is $n^4(6c^2 - 9c + 1)$ with $c = q/n$. The method therefore applies to all $q \le (\gamma - \varepsilon)n$ for $n \ge n_1(\varepsilon)$, where $\gamma = \frac{9-\sqrt{57}}{12} = 0.1208\ldots$ is the smaller root of $6c^2 - 9c + 1$. The exact thresholds beyond which $G(n,q) > 0$ holds for all larger n are $n_0(2) = 20$, $n_0(3) = 28$, $n_0(4) = 36$, $n_0(5) = 45$, $n_0(10) = 86$, $n_0(20) = 169$ and $n_0(50) = 417$.

Corollary 6.9. *Let $n \ge 40$. Every tight set $([n-1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ with $w_1, w_2 \ge n$ satisfies $\min(r_1, r_2) = 1$ or $\min(r_1, r_2) > n/10$.*

6.4 The census

Theorem 6.10 (Census)**.** *Let $8 \le n \le 140$, $1 \le r_1 < r_2 \le n-1$ and $n \le w_1 < w_2 \le 8n$. The set $([n-1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ is tight only in the following two cases:*

- *$n = 8$, $R = \{2, 3\}$, $W = \{11, 13\}$, giving $\{1, 4, 5, 6, 7, 11, 13\}$;*
- *$n = 74$, $R = \{70, 72\}$, $W = \{140, 144\}$, the example of Goddyn and Wong (Example 5.2 with $k = 2$).*

The theorem is the result of an exact computation, described in Section 7. The bound $w \le 8n$ is a pragmatic window: it contains the doubled speeds $2(n-2)$ of Example 5.2 with a fourfold margin, but no general bound on the inserted speeds of tight two-entry modifications is known (Question 8.2). Theorems 6.3 and 6.6 take over where the census stops, provided one removed speed lies in $[2, n/10]$; by Corollary 6.9, the remaining open case for $n \ge 40$ is that both removed speeds exceed $n/10$ or one of them equals 1.

7 Computations

All computations use exact rational arithmetic. Tightness of V, i.e. $\mathrm{LR}(V) = \frac{1}{n}$, is equivalent to the conjunction of two finite conditions: (i) the bands $\{t : \|vt\| \le \frac{1}{n}\}$, $v \in V$, cover the circle, which is decided by the finitely many endpoints $\frac{j}{v} \pm \frac{1}{vn}$; and (ii) some t has $\|vt\| \ge \frac{1}{n}$ for all $v \in V$, which is decided by evaluating $f_V(t) = \min_v \|vt\|$ at its finitely many breakpoints

verification	range	size	outcome
One-entry modifications (independent check of Theorem 1.2)			
middle range, effective bound	$r \le 30$, $n \le 6r$, $w \le \frac{4rI}{2s-I}$	21,789	tight only $(5,2,7)$, $(6,2,9)$
brute force, no effective bound	$3 \le r \le 8$, $w \le 30n$	—	none tight
Two-entry modifications			
census, all pairs R (Thm. 6.10)	$8 \le n \le 140$, $w \le 8n$	447,545	the two sets of Thm. 6.10
$R = \{2, r'\}$, deep window	$9 \le n \le 18$, $w \le 40n$	96,725	none tight
$R = \{2, 3\}$, deep window	$n = 8$, $64 < w_2 \le 320$	901	none covers
Three- and four-entry modifications (Question 8.3)			
$k = 3$, doubling type	$3 \le r \le 35$, $n \le 4r$	241,699	none tight
$k = 4$, all multipliers	$10 \le n \le 30$	12,646	none tight
Enumeration subtotal: 821,305 *cases,* 0 *unexpected tight sets.*			
Further exact checks			
$M(n)$ via (1)	$n \le 40{,}000$	—	$< 0.60\, n \log n + 52n$
primorial family	$p \le 19$	—	$m = \frac{q-1}{2}$ optimal
inequality (11)	$q \le 100$, $n \le 5000$	—	holds for $n \ge \max(40, 10q)$
deletion values	$n \le 30$, all r	432	consistent with Sec. 2
spectrum (Table 2)	$5 \le n \le 12$, $r = 2$, $w \le 8n$	484	smallest non-tight value as listed
Example 5.2, $n = 74$, 284	—	2	tight

Table 1: Exact-arithmetic verifications.

(the peaks $\frac{2k+1}{2v}$ and the crossings $\frac{k}{v \pm u}$). For two-entry modifications we use three independent implementations:

1. an *edge-reach census*: for each pair R, the inserted speeds are restricted by the necessary conditions of Lemma 6.2 type (a bad interval covering the left end a of a component has its centre index in $[wa - \frac{1}{n}, wa + \frac{1}{n}]$). The surviving pairs are screened in floating point, with bad intervals enlarged so that only non-covering pairs can be discarded, and every survivor is re-verified exactly;

2. a *union filter*, intersecting the bad intervals with the exact components of $U(n, R)$;

3. a *remaining-region method*, computing the set where both $\|w_1 t\| > \frac{1}{n}$ and $\|w_2 t\| > \frac{1}{n}$ and checking that it avoids $U(n, R)$. This method does not use the component structure.

They agree on all overlapping ranges. Table 1 lists the verifications. Sizes count candidate cases after the (necessary) edge-reach masks.

Remark 7.1 (Audit of the theoretical inequalities)**.** Beyond the searches, each inequality used in Section 3 was tested directly as an independent exact-arithmetic check, although Lemma 3.3 covers all $r \ge 3$ unconditionally: the equivalence of Lemma 3.1 with the exact integer-existence condition, and the sign claims of Lemma 3.2, over all 2,628 admissible triples (n, r, w) with $3 \le r \le 14$ and $w \le 4rI/(2s - I)$, $2r \le n - 1$ (no discrepancy, and no triple satisfied both windows); the implication $\omega(r) \ge 2 \Rightarrow 2s - I \ge 5$ and the threshold chain $2s - I \ge 5 \Rightarrow$

$4rI/(2s-I) < r(n-2)$, as well as the key inequality itself, for all $r \le 60$ and all admissible n (no violation). The estimate (8) was checked for all integers $101 \le x \le 2 \cdot 10^6$ in high-precision arithmetic against the exact primorials: the minimum of $\theta(x)/x$ over integer $x \ge 101$ on the range is 0.84976, attained at $x = 126$ (over real x it is $\theta(113)/127 = 0.84307$), both above 0.84. The thresholds of Lemma 7.2 were recomputed exactly and cross-checked against the census data.

7.1 Loneliness values just above $1/n$

By (5) and Theorem 2.11, every value $\mathrm{LR}([n-1]_{2\to w})$ lies in $[\frac{1}{n}, \frac{2}{D}]$, an interval that contains $(\frac{1}{n}, \frac{1}{n-1})$. By the theorem of Giri and Kravitz [17], the loneliness values of $(n-2)$-element sets, in particular $\frac{1}{n-1} = \mathrm{LR}([n-2])$, are accumulation points of the values of $(n-1)$-element sets. Table 2 lists, for $5 \le n \le 12$, the smallest value of $\mathrm{LR}([n-1]_{2\to w})$ exceeding $\frac{1}{n}$ over the searched range $n \le w \le W_0 = 8n$ (the census window of Section 7); each value is certified in exact rational arithmetic. The following lemma, applied at the table values, promotes the search to a complete classification over all $w \ge n$.

Lemma 7.2 (A uniform covering bound for $r = 2$)**.** *Let $n \ge 5$, let i_0 be the largest odd integer $\le n-1$ (so that $D = 2\lfloor n/2 \rfloor + 3 = i_0 + 4$), and let $0 < \lambda < \frac{2}{D}$. Put*

$$\ell(\lambda) = \frac{1/2 - \lambda}{i_0} - \frac{\lambda}{4} \quad (> 0).$$

If $w \ge n$ and $\mathrm{LR}([n-1]_{2\to w}) \le \lambda$, then $w \le 2\lambda/\ell(\lambda)$.

Proof. Let $J = \big(\frac{1}{2} + \frac{\lambda}{4},\ \frac{1}{2} + \frac{1/2-\lambda}{i_0}\big)$, which is nonempty, and write $t = \frac{1}{2} + \tau \in J$. For odd $i \le i_0$ we have $it \equiv \frac{1}{2} + i\tau \pmod 1$ with $0 < i\tau < \frac{1}{2} - \lambda$, so $\|it\| = \frac{1}{2} - i\tau > \lambda$. For even i with $4 \le i \le n-1$ we have $it \equiv i\tau \pmod 1$ and

$$\lambda < i\tau < \frac{(n-1)(1/2-\lambda)}{i_0} \le \frac{4}{3}\Big(\frac{1}{2} - \lambda\Big) < 1 - \lambda,$$

so again $\|it\| > \lambda$. Hence every speed of $[n-1]_{2\to w}$ other than w exceeds λ on J, and $\mathrm{LR} \le \lambda$ forces $\|wt\| \le \lambda$ on $\overline{J}$. As $\lambda < \frac{1}{2}$, the set $\{t : \|wt\| \le \lambda\}$ is a disjoint union of closed intervals of length $\frac{2\lambda}{w}$, one of which contains $\overline{J}$; thus $\ell(\lambda) \le \frac{2\lambda}{w}$. □

For $\lambda = \frac{1}{n}$ one has $\ell(\frac{1}{n}) = \Delta(n)$, and the lemma reduces to the length condition in the proof of Proposition 3.5. Since $2\lambda/\ell(\lambda)$ increases with λ, the lemma shows that $\mathrm{LR}([n-1]_{2\to w}) > v$ whenever $w > 2v/\ell(v)$. For the values $v = v_n$ of Table 2 one finds $2v_n/\ell(v_n) = 12, 20, 10, 14, 14, 18, 18, 22$ for $n = 5, \dots, 12$, which is well inside the searched range $w \le 8n$. Consequently Table 2 lists the smallest non-tight value of $\mathrm{LR}([n-1]_{2\to w})$ over *all* $w \ge n$; in particular, for $9 \le n \le 12$ no one-entry modification removing 2 has loneliness in $(\frac{1}{n}, \frac{1}{n-1})$.

For $n \le 8$ these values lie strictly between $\frac{1}{n}$ and $\frac{1}{n-1}$ and have the form $\frac{\sigma}{(n-1)\sigma+1}$ with $\sigma = 2$, as in Kravitz's conjecture [21]. For $9 \le n \le 12$ they exceed $\frac{1}{n-1}$. We record this as an observation only. Kravitz's conjecture is false in general [16], and our data concern a very special class of sets.

n	$1/n$	smallest value	at w	$< \frac{1}{n-1}$?	of the form $\frac{\sigma}{(n-1)\sigma+1}$?
5	1/5	2/9	5	yes	$\sigma = 2$
6	1/6	2/11	7	yes	$\sigma = 2$
7	1/7	2/13	9	yes	$\sigma = 2$
8	1/8	2/15	11	yes	$\sigma = 2$
9	1/9	2/15	11	no	—
10	1/10	2/17	13	no	—
11	1/11	2/17	13	no	—
12	1/12	2/19	15	no	—

Table 2: Smallest non-tight values of $\mathrm{LR}([n-1]_{2\to w})$: certified in exact arithmetic over the searched range $n \le w \le 8n$, and valid for all $w \ge n$ by Lemma 7.2. The equal values at $n = 10$ and $n = 11$ come from the distinct sets $[9]_{2\to 13}$ and $[10]_{2\to 13}$.

8 Open problems

Question 8.1. Classify the tight two-entry modifications $([n-1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$. By Corollary 6.9 and Theorem 6.10, the open cases are $n > 140$ or $w_2 > 8n$, with both $r_i > n/10$ or $\min r_i = 1$. When $r_1, r_2 > \frac{n-1}{2}$, Lemma 5.3 shows that each inserted speed covering $\frac{1}{r_i}$ is a multiple of r_i. Is every tight two-entry modification with $r_1, r_2 > \frac{n-1}{2}$ obtained from Theorem 5.1?

Question 8.2. Does a bound $\max V = O(n \log n)$, or any polynomial bound, hold for all tight modifications $([n-1] \setminus R) \cup W$ with $|R| = |W| \ge 2$? Corollary 5.4 gives such a bound under the Goddyn–Wong criterion, but for $|R| \ge 2$ tightness alone does not obviously imply the criterion for each inserted speed.

Question 8.3. Are there tight multi-entry modifications of $[n-1]$ that do not arise from Theorem 5.1, apart from sets such as $\{1, 4, 5, 6, 7, 11, 13\}$? The converse of Theorem 5.1 already fails in general [18, Sec. 3], but the searches of Table 1 found no such example.

Question 8.4. Is $\mathrm{LR}([n-1] \setminus \{r\}) > \frac{1}{n-1}$ for all $n \ge 5$ and all $r \le n-2$? By Theorem 2.9 and Corollary 2.10 this holds unless no integer in $[n+r, 2n-3]$ is coprime to r; in particular it can fail only for $r > n - 2 - 2\sqrt{n}$. The exact value is known for $r \le \frac{n-1}{2}$ by Liu's theorem [22, Thm. 18].

Question 8.5. In the notation of Appendix A: is $\Gamma(n) < \frac{1}{n+1}$ for every $n \ge 5$? What are the minimisers of $\Gamma(n)$?

A The shifted variant

For a vector $\mathbf{v} = (v_1, \dots, v_n)$ of distinct positive integers and starting positions $\mathbf{s} \in [0,1)^n$, put

$$\gamma(\mathbf{v}; \mathbf{s}) = \max_t \min_i \|s_i + v_i t\|, \qquad \gamma^{\min}(\mathbf{v}) = \min_{\mathbf{s}} \gamma(\mathbf{v}; \mathbf{s}), \qquad \Gamma(n) = \min_{|\mathbf{v}|=n} \gamma^{\min}(\mathbf{v}).$$

The *shifted Lonely Runner Conjecture* [5, Conj. 1] asserts $\gamma^{\min}(\mathbf{v}) \ge \frac{1}{n+1}$. It is closely related to the zonotopal reformulation of Henze and Malikiosis [19] and to the finite-checking results of Malikiosis, Santos and Schymura [23]. Alcántara, Criado and Santos proved it for four speeds

and found exactly three primitive tight instances, two of which are tight in the classical sense [1]. Blanco, Criado and Santos disproved it for $5 \le n \le 17$; for instance $\gamma^{\min}(1,2,3,4,5) = \frac{15}{94} < \frac{1}{6}$ [9] (the case $n = 6$ being $\gamma^{\min}(1,\dots,6) = \frac{9}{67} < \frac{1}{7}$). Writing $\Gamma(n) = 1/(n+1+E_n)$, Poliakova [25] proved $E_n \ge \lfloor n/287 \rfloor$, so that E_n grows linearly, and $E_n \ge 1$ for all $n \ge 95$. Together with the computations of [9], Question 8.5 is therefore open only for $18 \le n \le 94$.

Lemma A.1. *Deleting a speed cannot decrease $\gamma^{\min}$. In particular $\Gamma(n)$ is nonincreasing in n.*

Proof. Let $\mathbf{v}'$ arise from $\mathbf{v}$ by deleting v_j, and let $\mathbf{s}'$ be starting positions for $\mathbf{v}'$. Extend them by an arbitrary s_j. For each t the minimum over fewer coordinates is larger, so $\gamma(\mathbf{v}';\mathbf{s}') \ge \gamma(\mathbf{v};\mathbf{s}) \ge \gamma^{\min}(\mathbf{v})$; now minimise over $\mathbf{s}'$. For the second claim, append a speed to a minimiser of $\Gamma(n)$. □

The contrast with the classical case is sharp. Classically the baseline is tight for every n, and Theorems 1.1 and 5.1 supply infinitely many further tight sets. In the shifted setting the baseline fails from five speeds on, and no analogue of the Goddyn–Wong families is known. Lemma A.1 alone does not propagate failure from n to $n+1$, because the target $\frac{1}{n+1}$ decreases.

Data availability

The programs and run logs for Section 7 are available at `https://github.com/Qinrayn/lonely-runner-tight-II`, one standalone script in exact rational arithmetic per verification.

Declaration on the use of generative AI

The mathematical content of this paper originates from the author's research; the main theorems and the strategy of their proofs were conceived by the author. The AI assistant Kimi K3 (Moonshot AI) was used for drafting parts of the exposition and for language editing. All AI-assisted material was checked in detail by the author, who takes full responsibility for the content of the paper.

References

[1] D. Alcántara, F. Criado and F. Santos, *Covering radii of 3-zonotopes and the shifted Lonely Runner Conjecture*, Exp. Math., published online 2026; `doi:10.1080/10586458.2026.2651085`; arXiv:2506.13379.

[2] J. Allikvere, *Fourteen and fifteen lonely runners*, preprint, arXiv:2609.02604.

[3] J. Barajas and O. Serra, *The lonely runner with seven runners*, Electron. J. Combin. **15** (2008), #R48.

[4] M. Beck and S. Everett, *Lonely Runner relations*, preprint, arXiv:2609.06259.

[5] M. Beck, S. Hoşten and M. Schymura, *Lonely Runner polyhedra*, Integers **19** (2019), #A29.

[6] B. Bedert, *Riesz products and the Lonely Runner Conjecture: a wider gap of loneliness*, preprint, arXiv:2511.16636.

[7] U. Betke and J. M. Wills, *Untere Schranken für zwei diophantische Approximations-Funktionen*, Monatsh. Math. **76** (1972), 214–217.

[8] W. Bienia, L. Goddyn, P. Gvozdjak, A. Sebő and M. Tarsi, *Flows, view obstructions, and the lonely runner*, J. Combin. Theory Ser. B **72** (1998), 1–9.

[9] M. Blanco, F. Criado and F. Santos, *Coloopless zonotopes and counterexamples to the shifted Lonely Runner Conjecture*, preprint, arXiv:2603.24784.

[10] T. Bohman and F. Peng, *Coprime mappings and lonely runners*, Mathematika **68** (2022), 784–804.

[11] T. Bohman, R. Holzman and D. Kleitman, *Six lonely runners*, Electron. J. Combin. **8** (2001), no. 2, #R3.

[12] Y.-G. Chen, *On a conjecture in Diophantine approximations, II*, J. Number Theory **37** (1991), no. 2, 181–198.

[13] T. W. Cusick, *View-obstruction problems*, Aequationes Math. **9** (1973), 165–170.

[14] T. W. Cusick, *View-obstruction problems in n-dimensional geometry*, J. Combin. Theory Ser. A **16** (1974), 1–11.

[15] T. W. Cusick and C. Pomerance, *View-obstruction problems, III*, J. Number Theory **19** (1984), 131–139.

[16] H. T. Fan and A. Sun, *Amending the Lonely Runner Spectrum Conjecture*, Electron. J. Combin. **33** (2026), no. 1, #P1.38.

[17] V. Giri and N. Kravitz, *The structure of Lonely Runner spectra*, Math. Proc. Cambridge Philos. Soc. **180** (2026), no. 2, 343–361.

[18] L. Goddyn and E. B. Wong, *Tight instances of the lonely runner*, Integers **6** (2006), #A38.

[19] M. Henze and R. D. Malikiosis, *On the covering radius of lattice zonotopes and its relation to view-obstructions and the lonely runner conjecture*, Aequationes Math. **91** (2017), 331–352.

[20] H.-J. Kanold, *Über eine zahlentheoretische Funktion von Jacobsthal*, Math. Ann. **170** (1967), 314–326.

[21] N. Kravitz, *Barely lonely runners and very lonely runners: a refined approach to the Lonely Runner Problem*, Combin. Theory **1** (2021), #17.

[22] D. D.-F. Liu, *From rainbow to the lonely runner: a survey on coloring parameters of distance graphs*, Taiwanese J. Math. **12** (2008), no. 4, 851–871.

[23] R. D. Malikiosis, F. Santos and M. Schymura, *Linearly exponential checking is enough for the lonely runner conjecture and some of its variants*, Forum Math. Sigma **13** (2025), e164.

[24] G. Perarnau and O. Serra, *The Lonely Runner Conjecture turns 60*, Comput. Sci. Rev. **58** (2025), 100798.

[25] D. Poliakova, *More (shifted) runners, less loneliness*, preprint, arXiv:2609.23952.

[26] C. Pomerance, *Coprime matchings*, Integers **22** (2022), #A2.

[27] M. Rosenfeld, *The lonely runner conjecture holds for eight runners*, preprint, arXiv:2509.14111.

[28] M. Rosenfeld, *The lonely runner conjecture holds for nine runners*, preprint, arXiv:2512.01912.

[29] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94.

[30] T. Sungkawichai and T. Trakulthongchai, *Eleven, twelve, and thirteen lonely runners*, preprint, arXiv:2604.23906.

[31] T. Tao, *Some remarks on the lonely runner conjecture*, Contrib. Discrete Math. **13** (2018), no. 2, 1–31.

[32] T. Trakulthongchai, *Nine and ten lonely runners*, Electron. J. Combin. **33** (2026), no. 2, #P2.46.

[33] J. M. Wills, *Zwei Sätze über inhomogene diophantische Approximation von Irrationalzahlen*, Monatsh. Math. **71** (1967), 263–269.

[34] J. M. Wills, *Zur simultanen homogenen diophantischen Approximation. I*, Monatsh. Math. **72** (1968), 254–263.